Lecture 10: The Consumer Problem

Announcements

- Homework...

  good news: HW5 not due till next Thursday

  bad news: it's longer than average

  good news: HW6 will be optional

  bad news: you still need to know that material!

1 Where are we?

- So far, we've defined what it means for preferences or a choice rule to be rational,

  established conditions under which a utility representation exists,

  and discussed other restrictions we could put on preferences and what they mean for the utility representation

- Now, we move on to the numerical formulation of the consumer's problem

- First... any questions?

2 Budget Sets and the Consumer Problem

- Till now, we've considered $X \subseteq \mathbb{R}^k_+$,

  but considered a choice rule specifying a selection from any arbitrary subset $A \subseteq X$

- Now, we'll fix $X = \mathbb{R}^k_+$, but we'll choose A more thoughtfully,

  having the consumer choose specifically among the *set of consumption bundles they can afford*

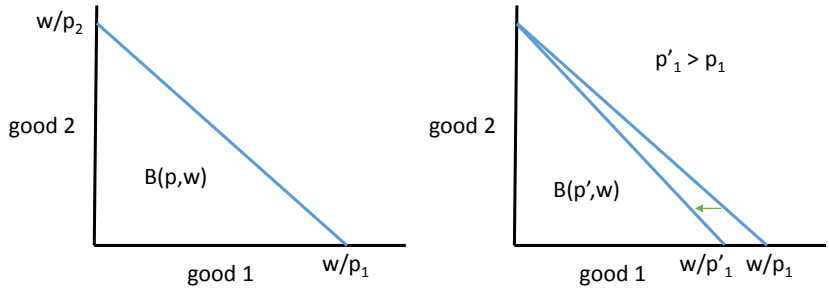
- To do this, we'll assume the consumer starts with some amount of wealth w,

  and faces some price vector p giving per-unit prices of each good

- we define the consumer's **budget set** as

$$B(p, w) \quad = \quad \left\{ x \in \mathbb{R}^k_+ \quad : \quad p \cdot x \leq w \right\}$$

- easiest to visualize in 2 dimensions:



- the consumer problem will then be

$$\max_{x \in B(p,w)} u(x)$$

  or

$$\max u(x) \quad s.t. \quad p \cdot x \leq w$$

- by formulating the problem this way, we're implicitly assuming...

  - **consumers are price-takers** – they take prices as fixed and non-negotiable (there's no bargaining, no searching for better prices, etc.)
  - **prices are linear** – each unit costs the same (no bulk discounts, etc.)
  - **goods are divisible** – we're letting $X = \mathbb{R}^k_+$, so any amount of each good can be consumed

- (we could relax the last two, but this is what we're starting with)

- also note there's just one opportunity to consume –

  we aren't letting the consumer save leftover money for future consumption (although...)

- Two things we can prove easily:
- **Proposition.** For any $\lambda > 0$, $B(\lambda p, \lambda w) = B(p, w)$, and if $p \gg 0$, $B(p, w)$ is compact.
  - For any positive λ, $p \cdot x \leq w$ if and only if $(\lambda p) \cdot x \leq \lambda w$.
    This is just saying the consumption bundles you can afford are the same if you count in dollars, or cents, or Malaysian ringgit.
  - For compactness: in Euclidean space, compact means closed and bounded
    Since B is defined by a weak inequality, it's closed
    And as long as all prices are strictly positive and w is finite, it's bounded
- **Proposition.** If u is continuous and $p \gg 0$, then the consumer problem

$$\max u(x) \quad s.t. \quad x \in B(p, w)$$

  has a solution.

  - Proof: a continuous function achieves a maximum on a compact space.
- We'll call the solution $x(p, w)$, that is,

$$x(p, w) \quad = \quad \arg \max_{x \in B(p,w)} u(x)$$

- We call this **Marshallian demand**, or **uncompensated demand** (you'll see why later)

- This is also the same as saying

$$x(p, w) \quad = \quad C(B(p, w); \succsim_u) \quad = \quad \{x \in B(p, w) \ : \ u(x) \geq u(y) \ \forall y \in B(p, w)\}$$

- recall a choice rule gave a *set* of optimal choices, not necessarily a singleton
- same here – $x(p, w)$ could be a set, not a single point
- so we can think of x as a correspondence:

$$x \quad : \quad \mathbb{R}^k_+ \times \mathbb{R}_+ \quad \rightrightarrows \quad \mathbb{R}^k_+$$

  mapping (p, w) to the set of optimal consumption bundles $x(p, w)$
  (the double-arrow means correspondence, not function,
  but under certain assumptions we can show it will be a function)

3 Some Properties of Marshallian Demand

- **Proposition.** Marshallian demand is homogeneous of degree zero: $x(\lambda p, \lambda w) = x(p, w)$ for any $p \gg 0$, $w > 0$, $\lambda > 0$.
  - Proof: we already showed $B(\lambda p, \lambda w) = B(p, w)$, so $x(\lambda p, \lambda w)$ and $x(p, w)$ are literally the solutions to the same optimization problem.
- Again, the problem is invariant to the unit of currency

- **Proposition [Walras' Law].** If preferences are locally non-satiated, then for any (p, w) and any $x \in x(p, w)$, $p \cdot x = w$.
  - Suppose not, i.e., suppose $p \cdot x < w$ for some $x \in x(p, w)$
  - If we make ε small enough, then for every y within ε of x, $p \cdot y \leq w$
    - specifically, let $\varepsilon = (w - p \cdot x)/\|p\|$
    - then
    $$p \cdot y \quad = \quad p \cdot (x + (y - x)) \quad = \quad p \cdot x + p \cdot (y - x)$$
    - since $a \cdot b = \|a\|\|b\| \cos\theta \leq \|a\|\|b\|$ (where θ is the angle between vectors a and b), this is
    $$\leq \quad p \cdot x + \|p\|\|y - x\| \quad \leq \quad p \cdot x + \|p\| \frac{w - p \cdot x}{\|p\|} \quad = \quad w$$
    so if $\|y - x\| \leq \varepsilon$, $p \cdot y \leq w$
  - but if preferences are LNS, then there must be some y within ε of x such that $y \succ x$
  - since we just showed that y was affordable, this contradicts x being the maximizer

- so if preferences are LNS, optimizing consumers always spend their entire budget;

  so if prefs are LNS, we can rewrite the consumer problem as

$$\max u(x) \quad s.t. \quad p \cdot x = w$$

- We said that $x(p, w)$ could be a set, i.e., the consumer problem could have multiple solutions
- **Proposition.**
  1. If preferences are convex, then for each p and w, $x(p, w)$ is a convex set
  2. If preferences are strictly convex, then for each p and w, $x(p, w)$ is a single point, i.e., x is a function.

  - Proof. $x(p, w)$ is a convex set if for any x and $x' \in x(p, w)$, $tx + (1 - t)x' \in x(p, w)$ as well. So let's show it.
  - If x and x' are both in $B(p, w)$, then $p \cdot x \leq w$ and $p \cdot x' \leq w$, so

$$p \cdot (tx + (1-t)x') \quad = \quad t(p \cdot x) + (1-t)(p \cdot x') \quad \leq \quad tw + (1-t)w \quad = \quad w$$

    so $tx + (1 - t)x'$ is in the budget set as well
  - And if x and x' are in $x(p, w)$, then for any other y in the budget set, $x \succsim y$ and $x' \succsim y$; if preferences are convex, this means $tx + (1-t)x' \succsim y$ as well, so $tx + (1-t)x' \in x(p, w)$

  - for the second part, if preferences are strictly convex, suppose there were two separate points x and x' in $x(p, w)$
  - since $x \succsim x'$ and $x' \succsim x'$, if preferences are strictly convex, for any $t \in (0, 1)$, $tx + (1 - t)x' \succ x'$
  - as we argued above, $tx+(1-t)x'$ is in the budget set, so this contradicts x' being optimal

- so now we know that Marshallian demand...
  - exists
  - is homogeneous of degree 0
  - exhausts the budget if preferences are LNS
  - is single-valued if preferences are strictly convex
- onward!

4 The Indirect Utility Function

- We stated the consumer problem as

$$\max u(x) \quad \text{subject to} \quad x \in B(p, w)$$

  and called the solution $x(p, w)$; now define $v(p, w)$ as the value function, i.e.,

$$v(p, w) \quad = \quad \max u(x) \quad \text{subject to} \quad x \in B(p, w)$$

- by definition, $v(p, w) = u(x)$ for any $x \in x(p, w)$ –

  v is the "utility value" of having wealth w and facing prices p

- note that $v : \mathbb{R}^k_{++} \times \mathbb{R}_+ \to \mathbb{R}$ is a function:

  if the CP has multiple solutions, they all give the same utility, so there's a unique max value

- v is called the *indirect utility function* (because it's a function of p and w, not consumption)

- Some properties of v:

- **Proposition.** Suppose u is a continuous utility function representing a LNS preference relation $\succsim$ on $\mathbb{R}^k_+$. Then $v(p, w)$ is...

  1. homogeneous of degree 0: $v(\lambda p, \lambda w) = v(p, w)$
  2. continuous on $\{(p, w) : p \gg 0, w \geq 0\}$
  3. nonincreasing in p, and strictly increasing in w
  4. quasi-convex, i.e., the set $\{(p, w) : v(p, w) \leq \bar{v}\}$ is a convex set for each $\bar{v}$

- I'll skip the formal proofs of most parts.

  – Part 1 is the usual – if you scale up prices and wealth, B is unchanged, so you're solving the same optimization problem.

  – Part 2, I'll skip the formal proof, but the intuition is this. If (p, w) is close to (p', w'), then the budget sets are close to being the same set; so if u is continuous, the utility you can achieve on one set is close to the utility you can achieve on the other.

  – Part 3, if p increases, the budget set shrinks, so you're maximizing the same function over a smaller set, so the value goes weakly down. If w goes up, then relative to what used to be your optimal point, you can afford strictly more, and can therefore afford any nearby bundle; by LNS, there's a nearby bundle that's strictly better, so you can afford at least one bundle better than your old optimum.

- Skipped in class, but here's the formal proof for part 4, that v is quasi-convex:
  - We need to show that for any $\bar{v}$, the set $\{(p, w) : v(p, w) \leq \bar{v}\}$ is a convex set
  - Or, if (p, w) and (p', w') are both in that set,
    so is $t(p, w) + (1 - t)(p', w')$, which we'll abbreviate (p^t, w^t)
  - so we need to show that if $v(p, w) \leq \bar{v}$ and $v(p', w') \leq \bar{v}$, then $v(p^t, w^t) \leq \bar{v}$ as well

  - To show this, note that for any $x \in B(p^t, w^t)$,
    x must also be in at least one of $B(p, w)$ or $B(p', w')$
  - If it weren't, then we'd know that both $p \cdot x > w$ and $p \cdot x' > w'$, which would mean

    $$t(p \cdot x) + (1 - t)(p \cdot x') > tw + (1 - t)w' \quad \longrightarrow \quad p \cdot x^t > w^t$$

    which would give a contradiction
  - So if $v(p, w)$ and $v(p', w')$ are both below $\bar{v}$,
    then any bundle with $u(x) > \bar{v}$ must be outside of both $B(p, w)$ and $B(p', w')$
    which we just said means it's outside of $B(p^t, w^t)$
  - so there's no way to achieve utility more than $\bar{v}$ in $B(p^t, w^t)$, so $v(p^t, w^t) \leq \bar{v}$.

5 Solving the Consumer Problem – Lagrangians

5.1 Introduction

- Next, we explore how we actually solve the consumer problem,

  which is important for two reasons:

  One, we need a way to actually solve the consumer problem and find Marshallian demand

  But two, it's a nice intro to how we might generally solve constrained optimization problems, which is kind of what economics is all about

- The fact that we're focusing on one problem makes things a tiny bit easier at one point –

  if you adapt these methods to a different constrained optimization problem,

  there's an additional restriction that has to hold,

  but this condition is always satisfied by the Consumer Problem

  (I'll mention that when we get there)

- One thing to note, though –

  even as we introduce powerful methods to solve problems like the Consumer Problem,

  you shouldn't think of these as a cookbook recipe

- That is, if you're given a problem to solve,

  don't blindly go looking for formulas to plug it into!

- Think about the logic of the problem – the "economics"

- If you're maximizing a utility function $x_1 + x_2$, you're probably going to be looking for a corner solution!

- If you're maximizing $\min\{x_1, x_2\}$, you're probably going to be consuming both goods in equal quantities!

- Try to think about the logic of what's going on first, before you try actually solving it!

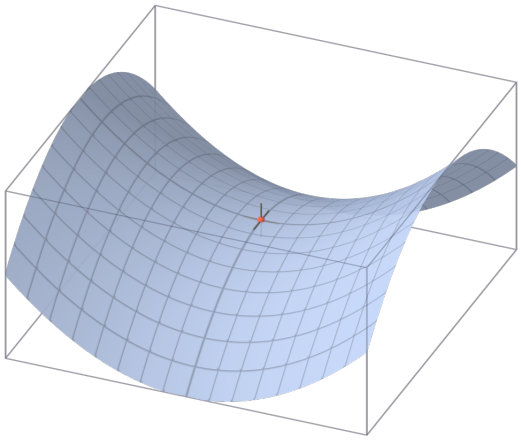
- That said, we are about to explore pretty general, powerful way to solve constrained maximization problems

5.2 Preliminaries: saddle points

- Consider a function $f : Y \times Z \to \mathbb{R}$, where $Y \subseteq \mathbb{R}^m$ and $Z \subseteq \mathbb{R}^n$
- A point (y^*, z^*) is a *saddle point* if it's a minimum with respect to one set but a maximum with respect to the other – for example, if

$$f(y, z^*) \quad \leq \quad f(y^*, z^*) \quad \leq \quad f(y^*, z)$$

  for any y and z
- That is, given z^*, y^* is the value of y that maximizes f, or y^* solves $\max_y f(y, z^*)$; but given y^*, z^* is the value of z that minimizes it, or z^* solves $\min_z f(y^*, z)$
- In two dimensions, a saddle point looks like this:



- If (y^*, z^*) is a saddle point of f, then y^* solves $\max_y f(y, z^*)$, and z^* solves

$$\min_z \max_y f(y, z)$$

  (If there was a different value of z, say z', with $\max_y f(y, z') < \max_y f(y, z^*)$, it would mean

$$f(y^*, z^*) \quad > \quad \max_y f(y, z') \quad \geq \quad f(y^*, z')$$

  which contradicts z^* being the minimizer given y^*.)
- So we can think of a saddle point (y^*, z^*) as a solution to the min-max problem

$$\min_z \max_y f(y, z)$$

5.3 More Preliminaries

- We'll be focusing on the case where preferences are represented by a utility function which is continuous and differentiable
- We'll be thinking of $X = \mathbb{R}^k_+$, but for technical reasons, we'll sometimes be thinking of choosing x out of all of $\mathbb{R}^k$, and imposing $x_i \geq 0$ as an added constraint (like the budget constraint)
- So we'll be thinking of the consumer problem as

$$\max_{x \in \mathbb{R}^k} u(x) \quad \text{subject to} \quad p \cdot x \leq w \quad \text{and} \quad x \geq 0$$

- (We could of course write this as

$$\max_{x \in \mathbb{R}^k_+} u(x) \quad \text{subject to} \quad p \cdot x \leq w$$

but the way we'll be solving it, it's more convenient to think of $\mathbb{R}^k$ as the space the problem is defined over, and $x_i \geq 0$ as additional constraints)

5.4 Lagrangians

- To solve the consumer problem – and constrained optimization problems in general – we'll introduce some auxiliary variables and functions
- Define some new variables $\lambda \in \mathbb{R}_+$ and $\mu = (\mu_1, \mu_2, \ldots, \mu_k) \in \mathbb{R}^k_+$, and define a function

$$\mathcal{L}(x, \lambda, \mu) \quad = \quad u(x) + \lambda(w - p \cdot x) + \sum_{i=1}^{k} \mu_i x_i$$

  on $\mathbb{R}^k \times \mathbb{R}^{k+1}_+$
- Note that we've written all our constraints as functions of x which must be ≥ 0,
  and then attached nonnegative variables to each of them
- The new variables are called Lagrange multipliers –
  note that there's one for each of the constraints on x in the consumer problem
- This function is called the Lagrangian
- (And note that while we're imposing nonnegativity on the Lagrange multipliers λ and μ, we're not imposing nonnegativity or feasibility on x –
  we're thinking of $\mathcal{L}$ as a function of (x, λ, μ), where x can take any value in $\mathbb{R}^k$,
  so $\mathcal{L}$ is a function on $\mathbb{R}^k \times \mathbb{R}_+ \times \mathbb{R}^k_+$.)

- We're interested in the saddle points of $\mathcal{L}$ – the points (x^*, λ^*, μ^*) such that

$$\mathcal{L}(x, \lambda^*, \mu^*) \quad \leq \quad \mathcal{L}(x^*, \lambda^*, \mu^*) \quad \leq \quad \mathcal{L}(x^*, \lambda, \mu)$$

  for any $x \in \mathbb{R}^k$ and any $(\lambda, \mu) \in \mathbb{R}^{k+1}_+$
- That is, we're interested in points where, given $\lambda = \lambda^*$ and $\mu = \mu^*$, x^* is maximizing $\mathcal{L}$;
  and given $x = x^*$, λ^* and μ^* are minimizing $\mathcal{L}$

- What's cool is, the saddle points of the Lagrangian are the solutions to the consumer problem:

Proposition 1. *Fix $(p, w) \gg 0$ (wealth and all prices are strictly positive).*

1. *If (x^*, λ^*, μ^*) is a saddle point of $\mathcal{L}$, then x^* solves the Consumer Problem.*
2. *If u is differentiable and concave and x^* solves the Consumer Problem, then there exist $\lambda^*, \mu^* \geq 0$ such that (x^*, λ^*, μ^*) is a saddle point of $\mathcal{L}$.*

Proof of Prop 1 part 1

Let's prove the first part – that a saddle point is a solution to the CP.

- Let B be the budget set – the subset of $\mathbb{R}^k$ satisfying $x \geq 0$ and $p \cdot x \leq w$
- If (x^*, λ^*, μ^*) is a saddle point, then $\mathcal{L}(x^*, \lambda, \mu) \geq \mathcal{L}(x^*, \lambda^*, \mu^*)$ for every $\lambda, \mu \geq 0$
- This in turn implies that $x^* \in B$
  - If $p \cdot x^* > w$, then $w - p \cdot x^*$ is negative;
    so we could decrease $\mathcal{L}$ by increasing λ above λ^*
  - And if there's a good i with $x_i < 0$,
    we could decrease $\mathcal{L}$ by increasing μ_i above μ_i^*
  - Either one would contradict the saddle condition, that

    $$\mathcal{L}(x^*, \lambda, \mu) \geq \mathcal{L}(x^*, \lambda^*, \mu^*)$$

    for all λ, μ (or that (λ^*, μ^*) minimize $\mathcal{L}(x^*, \cdot, \cdot)$)
- So at a saddle point, the constraints on x all hold, and x^* is therefore feasible

- Next, at a saddle point, a condition called **complementary slackness** holds:
  $\lambda^*(w - p \cdot x^*) = 0$ and $\mu_i^* x_i^* = 0$ for each i
- That is, for each constraint that doesn't hold with equality, the corresponding Lagrange multiplier must be 0
  - Suppose not
  - If $w - p \cdot x^* > 0$ and $\lambda^* > 0$, then lowering λ would lower $\mathcal{L}$,
    violating the saddle point condition
  - And if $x_i^* > 0$ and $\mu_i^* > 0$, then lowering μ_i would lower $\mathcal{L}$

- So, if (x^*, λ^*, μ^*) is a saddle point of $\mathcal{L}$,
  then x^* is feasible, and $\lambda^*(w - p \cdot x^*) = \mu^* \cdot x^* = 0$
- Complementary slackness is nice, because it means that at a saddle point,

$$\mathcal{L}(x^*, \lambda^*, \mu^*) \quad = \quad u(x^*) + \lambda^*(w - p \cdot x^*) + \mu^* \cdot x^* \quad = \quad u(x^*)$$

  (the value of the Lagrangian is just $u(x^*)$, because the other terms are 0)

- Now, remember what we're trying to show: that x^* is a solution to the consumer problem,
  i.e., that $u(x^*) \geq u(x)$ for every point x in the budget set
- So let's pick any other feasible point $x' \in B(p, w)$, and show $u(x) \geq u(x')$
- Since $\lambda^* \geq 0$ and $\mu^* \geq 0$, if x' is feasible, then

$$\mathcal{L}(x', \lambda^*, \mu^*) \quad = \quad u(x) + \lambda^*(w - p \cdot x') + \mu^* \cdot x' \quad \geq \quad u(x')$$

  because $w - p \cdot x' \geq 0$ and $x' \geq 0$ if it's feasible
- Now, the saddle point condition is that at (λ^*, μ^*), x^* maximizes $\mathcal{L}$, or

$$\mathcal{L}(x^*, \lambda^*, \mu^*) \quad \geq \quad \mathcal{L}(x', \lambda^*, \mu^*)$$

- We saw before that $\mathcal{L}(x^*, \lambda^*, \mu^*) = u(x^*)$,
  and $\mathcal{L}(x', \lambda^*, \mu^*) \geq u(x')$, so if we put it together,

$$u(x^*) \quad = \quad \mathcal{L}(x^*, \lambda^*, \mu^*) \quad \geq \quad \mathcal{L}(x', \lambda^*, \mu^*) \quad \geq \quad u(x')$$

- This holds for any x' that's feasible;
  and we know x^* is feasible;
  so x^* maximizes $u(x)$ over the feasible set,
  or x^* solves the consumer problem $\max_{x \in B} u(x)$

Proof of Prop 1 part 2

What about the second part – that any solution to the consumer problem gives a saddle point?

- Suppose u is differentiable and concave
- If $x^* = 0$ (the consumer optimally consumes nothing), the proof is different;[1]
  but suppose we have a solution $x^* \neq 0$ to the consumer problem
- We'll find values λ^* and μ^* that make (x^*, λ^*, μ^*) a saddle point of $\mathcal{L}$

- Specifically, since $x^* \neq 0$, we can pick a good j such that $x_j \neq 0$, and let

$$\lambda^* \quad = \quad \frac{1}{p_j} \frac{\partial u}{\partial x_j}(x^*)$$

- Then for each good i, set $\mu_i = 0$ if $x_i^* > 0$, and $\mu_i = \lambda p_i - \frac{\partial u}{\partial x_i}(x^*)$ if $x_i^* = 0$

- We'll show four things:
  1. $\lambda^* \geq 0$
  2. λ^* is uniquely defined, that is, it doesn't depend on which good we chose
     (or $\frac{1}{p_j} \frac{\partial u}{\partial x_j}(x^*)$ is the same for each j with $x_j^* > 0$)
  3. $\mu^* \geq 0$
  4. (x^*, λ^*, μ^*) is a saddle point of $\mathcal{L}$

- First, $\lambda^* = \frac{1}{p_j} \frac{\partial u}{\partial x_j}(x^*)$ must be weakly positive.
  Since prices are positive, if $\lambda^* < 0$, that would mean u was strictly decreasing in x_j at x^*;
  but since $x_j^* > 0$, that would mean you could increase utility by consuming less of good j,
  which would still have satisfied the budget constraint,
  so x^* couldn't have been a solution to the consumer problem.

[1] Specifically, if $x^* = 0$ solves the consumer problem, then $\frac{\partial u}{\partial x_i}(0) \leq 0$ for each i. In this case, setting $\lambda^* = 0$ and $\mu_i^* = -\frac{\partial u}{\partial x_i}(0)$ give a saddle point; you can try proving this on your own.

- Next, λ^* is uniquely defined
- Suppose not, i.e., suppose there were two goods, i and j, with $x_i^*, x_j^* > 0$, and

$$\frac{1}{p_i}\frac{\partial u}{\partial x_i}(x^*) \quad > \quad \frac{1}{p_j}\frac{\partial u}{\partial x_j}(x^*)$$

- Then it's easy to show that by consuming a little less of good j,
  and using the money saved to buy more of good i,
  the consumer would get strictly higher utility
  - If you spend ϵ less on good j, you get $\frac{\epsilon}{p_j}$ less of it;
    for ϵ small, this reduces utility by
    $$\frac{\epsilon}{p_j}\frac{\partial u}{\partial x_j}(x^*)$$
  - If you then spend ϵ more on good i, you get $\frac{\epsilon}{p_i}$ more, giving
    $$\frac{\epsilon}{p_i}\frac{\partial u}{\partial x_i}(x^*)$$
    more utility
  - So if
    $$\frac{\epsilon}{p_i}\frac{\partial u}{\partial x_i}(x^*) \quad > \quad \frac{\epsilon}{p_j}\frac{\partial u}{\partial x_j}(x^*)$$
    and $x_j^* > 0$, this would increase utility –
    so x^* wouldn't be a solution to the consumer problem
- (This is the observation that if you're choosing optimally,
  your marginal rate of substitution must equal the price ratio)

- Third, $\mu_i^* \geq 0$ for each i
- If $x_i^* > 0$, we set $\mu_i^* = 0$, so we only need to show it for $x_i^* = 0$
- So suppose $x_i^* = 0$, and suppose that $\mu_i^* < 0$ as defined
- Then

$$0 \quad > \quad \lambda^* p_i - \frac{\partial u}{\partial x_i}(x^*) \qquad \longrightarrow \qquad \lambda^* \quad < \quad \frac{1}{p_i}\frac{\partial u}{\partial x_i}(x^*)$$

- But going back to how we defined λ^*, this means

$$\frac{1}{p_j}\frac{\partial u}{\partial x_j}(x^*) \quad < \quad \frac{1}{p_i}\frac{\partial u}{\partial x_i}(x^*)$$

  for some (i, j) with $x_j^* > 0 = x_i^*$
- But then we could gain by spending ϵ less on good j and ϵ more on good i
- Since that can't happen, it must be that $\mu_i^* \geq 0$

- What's left, then, is to show that these particular multipliers give a saddle point
- First, let's show that

$$\mathcal{L}(x, \lambda^*, \mu^*) \quad \leq \quad \mathcal{L}(x^*, \lambda^*, \mu^*)$$

  for all $x \in \mathbb{R}^k$ (feasible or not)
- Note that

$$\frac{\partial \mathcal{L}}{\partial x_i}(x^*, \lambda^*, \mu^*) \quad = \quad \frac{\partial u}{\partial x_i} - \lambda p_i + \mu_i \Big|_{(x,\lambda,\mu)=(x^*,\lambda^*,\mu^*)} \quad = \quad 0$$

  by the definition of λ^* and μ^*

  (If $x_i^* > 0$, then $\lambda^* = \frac{\partial u}{\partial x_i}(x^*)$ and $\mu_i^* = 0$;

  and if $x_i^* = 0$, then μ_i^* was defined to make $\frac{\partial \mathcal{L}}{\partial x_i} = 0$ hold.)
- So holding fixed $\lambda = \lambda^*$ and $\mu = \mu^*$, $\mathcal{L}$ is flat in x at x^*
- But since u is concave, and $\mathcal{L}$ is just u plus some linear terms, $\mathcal{L}$ is concave in x, so this means $\mathcal{L}$ is maximized at x^*

THIS IS WHERE WE RAN OUT OF TIME!

the missing part of the proof, which we'll start with on Thursday:

- Second, we need to show

$$\mathcal{L}(x^*, \lambda, \mu) \quad \geq \quad \mathcal{L}(x^*, \lambda^*, \mu^*)$$

for all $\lambda, \mu \geq 0$

- Note that the multipliers we defined satisfy complementary slackness:

if $p \cdot x^* < w$ is optimal,

it must be that $\frac{\partial u}{\partial x_i}(x^*) \leq 0$ for all i, and $= 0$ for all i with $x_i^* > 0$,

so $\lambda^* = 0$;

if not, $p \cdot x^* = w$, so $\lambda^*(w - p \cdot x^*) = 0$ either way

- And we set $\mu_i^* = 0$ for any i with $x_i^* = 0$, so $\lambda^*(w - p \cdot x^*) = 0$ and $\mu^* \cdot x^* = 0$

- Now at $x = x^*$, $\mathcal{L}$ is linear in λ, with slope $w - p \cdot x^*$

- If $w - p \cdot x^* > 0$, then $\mathcal{L}$ is increasing in λ, but $\lambda^* = 0$, so $\mathcal{L}$ is being minimized;

and if $w - p \cdot x^* = 0$, then $\mathcal{L}$ is constant in λ

- Similarly, if $x_i^* > 0$, then $\mathcal{L}$ is increasing in μ_i, but $\mu_i^* = 0$, so $\mathcal{L}$ is being minimized;

and if $x_i^* = 0$, then $\mathcal{L}$ is constant in μ_i

- So that's it – if u is differentiable and concave and $(p, w) \gg 0$,

then for any solution to the consumer problem, there's a saddle point of $\mathcal{L}$